Predicting County-Level Homeownership Rates in the United States Using Machine Learning and Historical ACS Data

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***Abstract*—This study predicts county-level homeownership rates in the United States using socioeconomic, demographic, and housing data from the American Community Survey (ACS) 5-year estimates. A distributed analytics pipeline was implemented using Databricks to process multi-year county-level data. Three machine learning models, Random Forest, Gradient Boosting (XGBoost), and Neural Networks were evaluated. Random Forest achieved the strongest predictive performance. Historical features, including prior-year homeownership and income changes, significantly improved predictive accuracy. Results demonstrate the value of nonlinear, ensemble-based approaches for modelling complex housing dynamics and provide insights for housing policy and regional planning.**

***Index Terms*—Homeownership, Random Forest, XGBoost, Neural Networks, ACS**

I. Introduction

Homeownership is a central indicator of economic stability, wealth accumulation, and community development. County-level homeownership patterns reflect interactions among income, education, labour markets, housing supply, demographic composition, and migration dynamics. Because these relationships are nonlinear and interdependent, traditional linear regression models may not fully capture the complexity of housing outcomes.

Using U.S. Census data and specifically the API provided for The American Community Survey (ACS) to detail county-level socioeconomic and housing data across time. Combining with distributed machine learning frameworks such as Databricks enables scalable modelling of housing patterns within the U.S.

This study tries to predict county-level homeownership rates using both current and historical ACS indicators. By incorporating historical features (ACS 2019) and creating change features from more recent data (ACS 2022), the analysis evaluates whether nonlinear ensemble models outperform neural networks in tabular socioeconomic datasets.

The project implementation follows three key parts. (1) data acquisition and ingestion from the ACS provided API, (2) exploratory data analysis and feature engineering to create data suitable for analysis. (3) Modelling using three supervised algorithms to evaluate performance and to identify key predictors of home ownership.

A. Research Questions

- Can county-level homeownership rates in the United States be predicted using a combination of current and historical socioeconomic, demographic, and housing characteristics?
- Which socioeconomic and housing factors are most predictive of homeownership at the county level?
- How effectively can different machine learning approaches model county-level homeownership rates?

II. Literature Review

Forecasting housing-related outcomes is inherently complex due to nonlinear interactions between demographic and economic variables. Traditional regression approaches often impose additive assumptions that may overlook structural interactions.

Robinson and Dilkina [2018] demonstrate that machine learning models outperform conventional migration models by capturing nonlinear interactions among high-dimensional predictors. Their work highlights the importance of flexible modelling frameworks for socio-economic forecasting.

Similarly, Mora-Garcia et al. [2022] show that ensemble learning methods—such as Random Forest and Gradient Boosting—outperform linear models in housing price prediction. They emphasize the importance of feature engineering, hyperparameter tuning, and overfitting control.

Farahi and Jiao [2024] apply interpretable machine learning to study racial disparities in homeownership, revealing nonlinear and geographically heterogeneous effects. Their findings reinforce the importance of flexible models capable of uncovering structural complexity.

Collectively, this literature supports three conclusions:

- Housing outcomes involve nonlinear predictors
- Ensemble tree-based models perform well on tabular data
- Historical data improves predictive performance

Building on these insights, this study evaluates Neural Networks, Random Forest, and XGBoost for predicting county-level homeownership rates using current and historical ACS data.

III. Data Acquisition and Ingestion

A. Data Source

I used The American Community Survey (ACS) provided by Census U.S as it gives yearly data on topics such as U.S. Census Bureau [2019] “social, economic, housing, and

demographic characteristics of the U.S. population." The available API provided consistent yearly data meaning feature engineering was more accessible.

I have chosen the data sets below for comparison

- 2019 ACS 5-year estimates
- 2022 ACS 5-year estimates

These datasets contain observations for 3,144 U.S. counties.

B. Data Retrieval Process

The Census API enabled automated access to ACS tables. API calls were implemented using the Python request library. The API returned data in JSON format which was converted into Pandas DataFrames.

```
url = "https://api.census.gov/data/2022/acs/acs5"
params = {

    "get": ",".join([
        "NAME","B07001_001E","B07001_002E","B19013_001E",

"B15003_001E","B15003_022E","B15003_023E","B15003_024E","B15003_025E",
        "B17001_001E","B17001_002E","B23025_003E","B23025_005E",
        "B25077_001E","B25003_001E","B25003_002E",
        "B25032_002E","B25032_003E","B25032_004E","B25032_005E",
        "B25032_006E","B25032_007E","B25032_008E","B25032_009E",
        "B25064_001E","B01001_001E","B01001_002E"
    ]),
    "for":"county:*"
}


r = requests.get(url, params=params)
data = r.json()
df = pd.DataFrame(data[1:], columns=data[0])
```

Fig. 1. API request implementation used to retrieve ACS variables

A set of ACS variables representing demographic, socioeconomic, and housing indicators were selected and added as query parameters in the API call. The dataset included variables related to income, poverty, education, labour force participation, housing tenure, migration, and housing values. The US Census website provides the variables are their mappings U.S. Census Bureau [2022]

Once the data was retrieved a numeric conversion was necessary to convert string values. Invalid records were also removed to avoid zero divisions later in calculations.

```
# Numeric conversion
geo_cols = ["NAME","state","county"]
num_cols = [c for c in df.columns if c not in geo_cols]
df[num_cols] = df[num_cols].apply(pd.to_numeric, errors="coerce")

# Filter zeros for safe divisions later. keeps only rows where all variables are greater than 0.
df = df[
    (df["B07001_001E"]>0)& #Total Population by Migration
    (df["B15003_001E"]>0)& #Total Population 25+
    (df["B17001_001E"]>0)& #Total Population in Poverty
    (df["B23025_003E"]>0)& #Total Labor Force
    (df["B25003_001E"]>0)& #Total Housing Units
    (df["B01001_001E"]>0) #Total Population
]
```

Fig. 2. Numeric Conversion from Strings

C. Dataset Integration

The 2019 dataset was merged with 2022 required a unique Identifier. A National Institute of Standards and Technology [YEAR] FIPS (Federal Information Processing System) identifier was created by combining the state and county codes this unique geo-identifier ensures that data aligns with the correct county observations.

```
# Unique county identifier (Federal Information System (FIPS) codes for
geographic processing standards)
df["county_fips"] = df["state"] + df["county"]
```

Fig. 3. Creation of the FIPS county identifier

IV. Data Description and Exploratory Analysis

A. Data Source

Data was obtained from the U.S. Census Bureau's American Community Survey (ACS) 5-year estimates for 2019 and 2022. The dataset includes 3,144 U.S. counties and integrates socioeconomic, demographic, and housing characteristics.

B. Variables

The dataset includes 36 numeric features grouped as follows.

Socioeconomic Indicators

- Log median household income
- Poverty rate
- Bachelor's degree attainment
- Unemployment rate

Socioeconomic variables capture the economic conditions that influence households' ability to purchase homes. Median household income reflects purchasing power, while poverty and unemployment rates capture economic vulnerability that may limit access to homeownership.

Housing Characteristics

- Log median home value
- Median gross rent
- Single-family owner occupied housing
- Percent owner occupied housing

Housing market variables measure the affordability and structure of local housing supply. Median home value and rent represent local price levels, while the proportion of single-family owner-occupied housing reflects the availability of housing types associated with ownership.

Demographics

- Male population share
- Population density proxy

Demographic variables capture population characteristics that influence housing demand and settlement patterns.

Historical Features

- Prior-year homeownership rate
- Income change
- Home value change
- Migration rate

Historical indicators derived from the 2019 dataset capture temporal persistence in housing markets. Prior homeownership rates, income levels, and housing values allow models to learn structural trends and lagged effects.

Interaction & Polynomial Features

- Income × home value
- Poverty × home value
- Education × home value
- Density × income
- Squared log income
- Squared log home value

Log transformations were applied to income and home value to reduce skewness. Missing values were minimal and addressed via median imputation.

C. Descriptive Patterns

Average homeownership increased from 71.6% in 2019 to 72.7% in 2022.

Median home values rose substantially during the same period.

D. Descriptive Analysis

Descriptive statistics were calculated for all numerical variables to assess central tendencies, dispersion, and potential data quality issues. Statistics included the mean, median, standard deviation, quartiles, skewness, and counts of missing values.

Remaining missing values were addressed using median imputation, which is robust to skewed distributions.

Descriptive Statistics of Numeric Features:

	count	mean	std	min	25%	50%	75%	max	skew	missing
homeownership_rate	3144.0	0.727	0.086	0.000	0.684	0.740	0.785	0.970	−1.332	0
log_income	3144.0	11.024	0.251	9.793	10.870	11.018	11.165	12.046	0.194	0
log_homevalue	3144.0	12.051	0.486	10.609	11.704	11.992	12.344	14.181	0.691	0
poverty_rate	3144.0	0.144	0.060	0.016	0.100	0.134	0.175	0.558	1.122	0
unemployment_rate	3144.0	0.050	0.025	0.000	0.034	0.047	0.060	0.320	1.858	0
pct_bachelors_plus	3144.0	0.235	0.101	0.000	0.165	0.209	0.279	0.789	1.356	0
male_share	3144.0	0.505	0.025	0.405	0.492	0.500	0.510	0.691	2.510	0
population_density_proxy	3144.0	1.000	3.170	0.000	0.103	0.245	0.646	94.356	13.181	0
median_gross_rent	3144.0	899.940	279.923	297.000	726.000	824.000	984.000	2805.000	1.840	0

Fig. 4. Descriptive Statistics

E. Income Distribution and Log Transformation

Median household income displayed significant positive skewness, reflecting large disparities between high-income metropolitan counties and lower-income rural areas. A logarithmic transformation (log(1 + income)) was applied to reduce skewness and stabilise variance, which improves model performance and interpretability.

Median housing values also showed strong right skewness.

```
income_log = np.log1p(income_orig)
```

Fig. 5. Code snippet for log transform

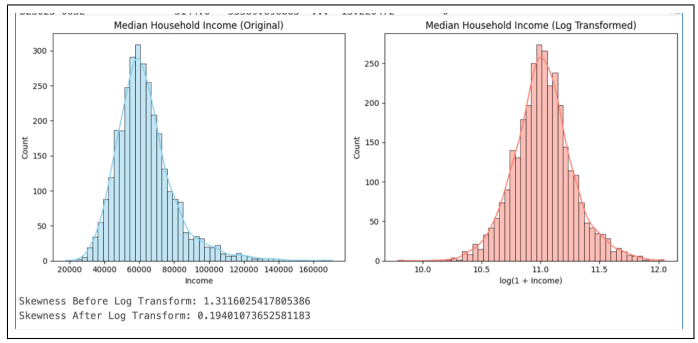


Fig. 6. Distribution of median household income before and after log transformation

F. Distribution and Transformations

Homeownership rates across counties cluster between 60% and 80%, with lower values observed in urban counties where rental housing is more prevalent. Educational attainment displays moderate variation across counties, with higher concentrations in metropolitan regions.

G. Correlation Analysis

A correlation matrix was computed to examine relationships among predictors. Strong positive relationships were observed between income, home value, and educational attainment, while poverty and unemployment exhibited negative correlations with homeownership.

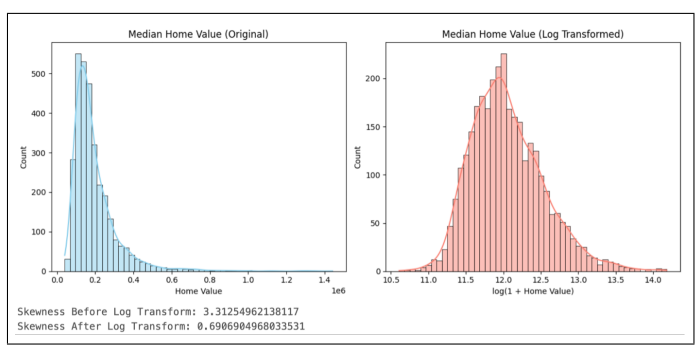


Fig. 7. Distribution of median home value income before and after log transformation

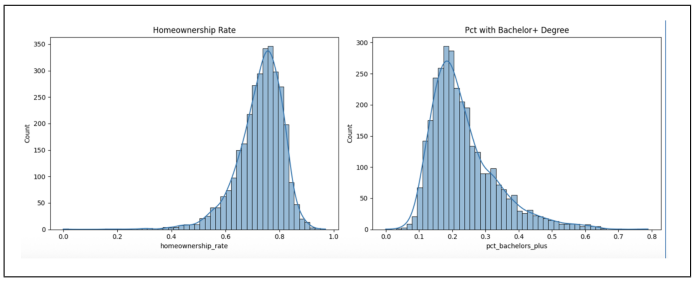


Fig. 8. Distribution and transformation

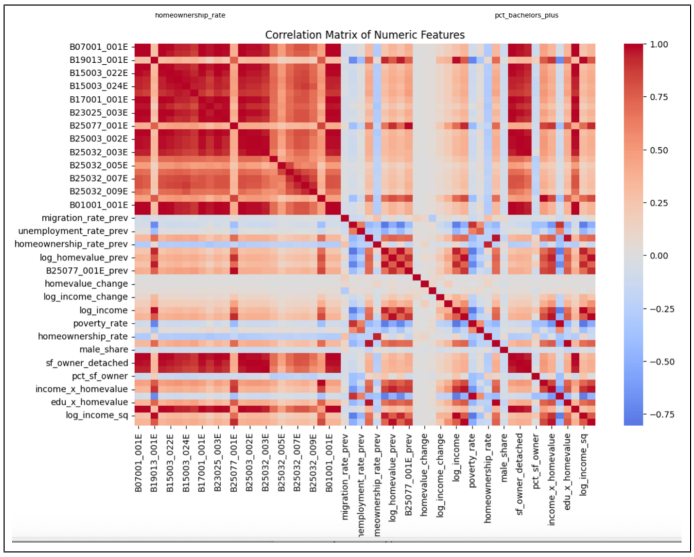


Fig. 9. Correlation matrix of key predictors

V. Methodology

A. Modelling Framework

Multiple models were implemented to compare predictive performance across different machine learning paradigms.

- Neural Network
- Random Forest Regressor
- Gradient Boosting Regressor (XGBoost)

The dataset was split into 80% training and 20% testing sets.

Model evaluation metrics:

- Mean Squared Error (MSE)
- Mean Absolute Error (MAE)
- R^2

B. Neural Network

A fully connected forward neural network was constructed:

- Input layer: 36 standardized features
- Hidden Layer 1: 128 neurons, ReLU activation, 30% dropout
- Hidden Layer 2: 64 neurons, ReLU activation, 20% dropout
- Output layer: 1 neuron

Training settings:

- Loss function: MSE
- Optimizer: Adam
- Early stopping
- Batch size: 128

C. Random Forest

The Random Forest model consisted of:

- 500 decision trees
- Bootstrap sampling
- Random feature selection

D. Gradient Boosting (XGBoost)

The XGBoost model used:

- 800 estimators
- Maximum depth: 6
- Learning rate: 0.03
- Subsampling: 0.8

VI. Results

TABLE I
Model performance comparison

Model	MSE	MAE	R^2
Neural Network	0.001684	0.030784	0.770
Random Forest	0.000879	0.021043	0.880
Gradient Boosting	0.000973	0.021551	0.867

A. Model Comparison

Random Forest achieved the highest predictive performance ($R^2 = 0.880$), followed closely by XGBoost ($R^2 = 0.867$). The Neural Network performed substantially worse ($R^2 = 0.770$).

B. Feature Importance

Feature importance analysis indicates that the strongest predictors include:

- Prior-year homeownership rate
- Median household income
- Median home value
- Income change
- Education attainment

Historical features substantially improved model performance, confirming the importance of temporal persistence in housing dynamics.

VII. Discussion

An analysis of the direction of county-level changes in homeownership rates between 2019 and 2022 reveals that the majority of counties experienced increases over this period. Specifically, 2,174 counties showed an increase in homeownership rates, while 969 counties experienced a decline and only one county exhibited no change. These results indicate that the overall upward trend observed in the descriptive statistics was broadly distributed across the United States, although a substantial number of counties still experienced declining homeownership. The presence of both increases

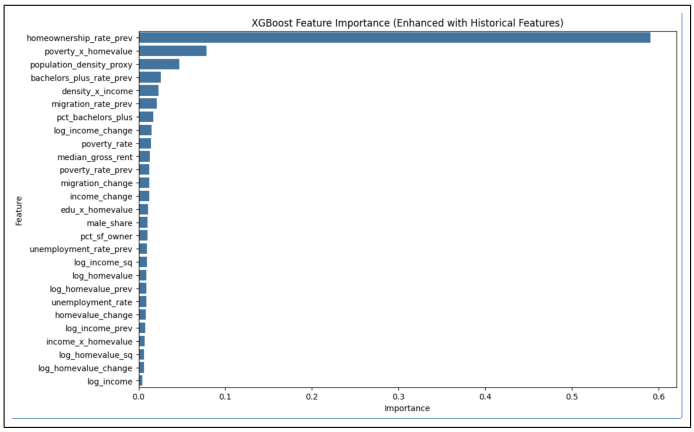
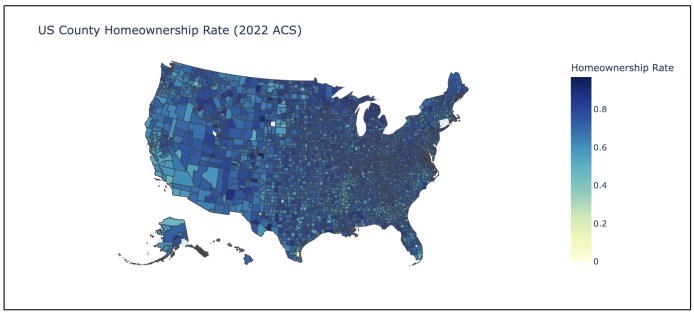
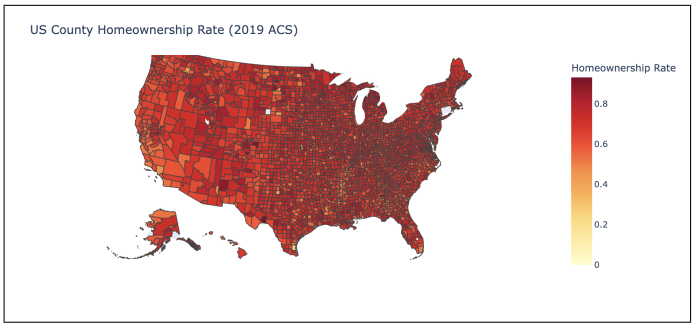


Fig. 10. Random Forest feature importance

and decreases highlights the heterogeneous nature of housing market dynamics, suggesting that local economic conditions, demographic trends, and housing supply constraints likely contributed to differing outcomes across counties.





Based on the results of this study, it is clear that county-level homeownership rates in the United States can be effectively predicted using a combination of current and historical socioeconomic, demographic, and housing characteristics. All three machine learning models evaluated—Neural Networks, Random Forest, and Gradient Boosting (XGBoost)—demonstrated strong predictive performance, with R^2 values ranging from 0.770 to 0.880, indicating that these features capture substantial variance in homeownership outcomes across counties. Among the predictor variables, prior-year homeownership rates, the interaction between poverty levels and housing prices, population density, and educational attainment emerged

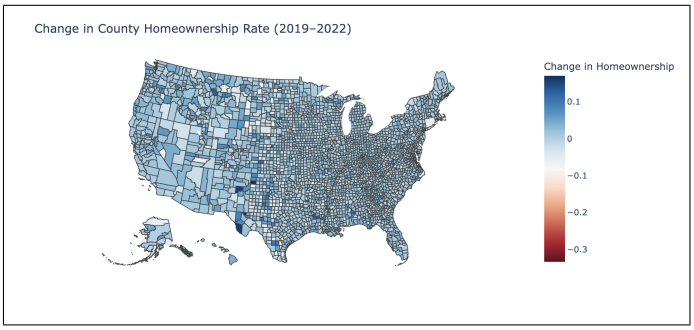


Fig. 11. Evolution of homeownership

as the most influential factors. These results highlight the importance of both historical persistence and broader socioeconomic conditions in shaping county-level homeownership patterns. In terms of model performance, Random Forest achieved the highest predictive accuracy ($R^2 = 0.880$), followed closely by Gradient Boosting ($R^2 = 0.8673$) and Neural Networks ($R^2 = 0.770$). These findings suggest that ensemble tree-based models are particularly well-suited for tabular socioeconomic data with nonlinear interactions, while neural networks can still provide strong predictions but may be constrained by dataset size and feature dimensionality.

In general, the results demonstrate that integrating historical features into predictive models not only improves accuracy but also provides valuable insights into the drivers of county-level homeownership, offering practical implications for policymakers and regional planners interested in housing stability and economic development. However, several limitations exist:

- Only two time periods were analysed
- Spatial dependencies were not modelled
- ACS estimates contain sampling variability

VIII. Conclusion

This study developed a machine learning framework to predict U.S. county-level homeownership rates using ACS data. The results demonstrate that county-level homeownership can be predicted with high accuracy using socioeconomic, demographic, and housing characteristics. Among the evaluated models, Random Forest achieved the strongest performance ($R^2 = 0.880$), outperforming both neural network and XGBoost models and highlighting the effectiveness of ensemble tree methods for structured socioeconomic data.

Feature importance analysis identified prior-year homeownership rates as the most influential predictor of homeownership outcomes. These findings provide insight into the structural drivers of homeownership and offer actionable information for policymakers and housing planners.

Future research could extend this work by incorporating spatial econometric modelling to account for geographic dependencies and by leveraging longer temporal datasets to better capture dynamic housing trends over time.

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